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Paleomagnetic Study on the Dike Swarm in the Oga Peninsula, Northeast Honshu Island

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A paleomagnetic study has been made on 14 dikes emplaced at two formations in the Oga Peninsula. Stable remanences of seven dikes are normally magnetized, six are reversely magnetized, and one has an intermediate direction. No significant difference in the directions is observed between the normally and the reversely magnetized rocks and between the intruded formations. The veins of dikes trend northeast with the standard deviation of 4°. Mean inclination and declination of remanence for 13 dikes excluding an intermediate one are calculated to be 58.3° and −2.8°, respectively. The pole position corresponding to the mean field direction is close to that observed at the Middle Miocene formations in the peninsula and in the northern part of NE Honshu Island. Dike rocks indicate the bimodal volcanism characteristic in Middle Miocene time. The direction of stable remanences and the bimodal distribution suggest that the dikes intruded in Middle Miocene time. The dominant orientation of strikes in the Oga dikes represents one of the general trend of the Middle to Late Miocene stress field after the tectonic rotation of NE Honshu Island had completed.

1. Introduction

During the last few years paleomagnetic studies have been intensively made on Cenozoic rocks in the Japanese Islands. Tectonic movements of the islands have been discussed based on their results. Major paleomagnetic results of these works were recently reviewed by TOSHA and KONO (1985) and HIROOKA (1983). Topics on Cenozoic time are summarized as follows: the directions of the natural remanent magnetizations (NRMs) in pre-Miocene rocks were systematically different from those at present in northeast (NE) and southwest (SW) Honshu Islands. Therefore it is inferred that NE and SW Honshu Islands have undergone counterclockwise and clockwise rotation, respectively (e.g., OTOFUJI and MATSUDA, 1983; OTOFUJI *et al.*, 1985; TOSHA and HAMANO, 1986).

The analysis of paleo-stress field is one of the most useful methods to discuss the tectonics, which is derived from the directions of strikes in dike swarms and faults and those of veins in ore deposits. Besides as a stress indicator most of the dikes are composed of igneous rocks and their remanent magnetizations are usually intensive and less disturbable by some late tectonic processes. Hence, dikes are also useful for paleomagnetic works. If we could measure both the strikes of dikes and the directions of their remanences, we could obtain the relation between the paleo-stress field and

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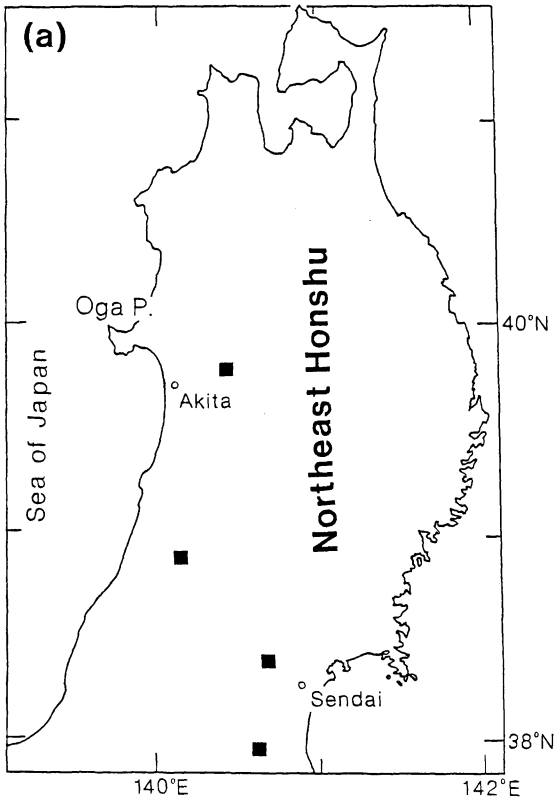


Fig. 1. (a) The Oga Peninsula is situated on the Sea of Japan side at about forty degrees north in latitude near the Akita City. Solid squares indicate the localities where dikes with the same dominant strikes as the Oga dikes are found (SATO *et al.*, 1982). (b) Sampling sites of the Oga dike swarm on a geological map after TAKAYASU and MATOBA (1976). Capital letters of AD and MD indicate the dikes penetrating into the Akashima Formation and the Monzen Formation, respectively.

the block movement, and possibly estimate the causality relation. However, paleomagnetism in dikes has disadvantages in that the intrusive ages are hardly determined by stratigraphic estimation and that no bedding corrections are performed. We collected samples of dikes with almost vertical dips, under the assumption that dikes were intruded almost vertically, and performed no stratigraphic correction. In this paper we will present paleomagnetic results of the dikes in NE Honshu Island and discuss their tectonic implication associated with the paleostress field.

2. Geological Setting

The Oga Peninsula is situated on the Sea of Japan side in Akita Prefecture, NE Honshu Island (Fig. 1(a)). Volcanic and sedimentary rocks of the Cenozoic are observed there. In the western part of the peninsula sequence of volcanic rocks are well exposed. In the older sequence volcanic rocks are dominant which are named in ascending order as follows: basement (about 62 Ma BP), the Akashima Formation

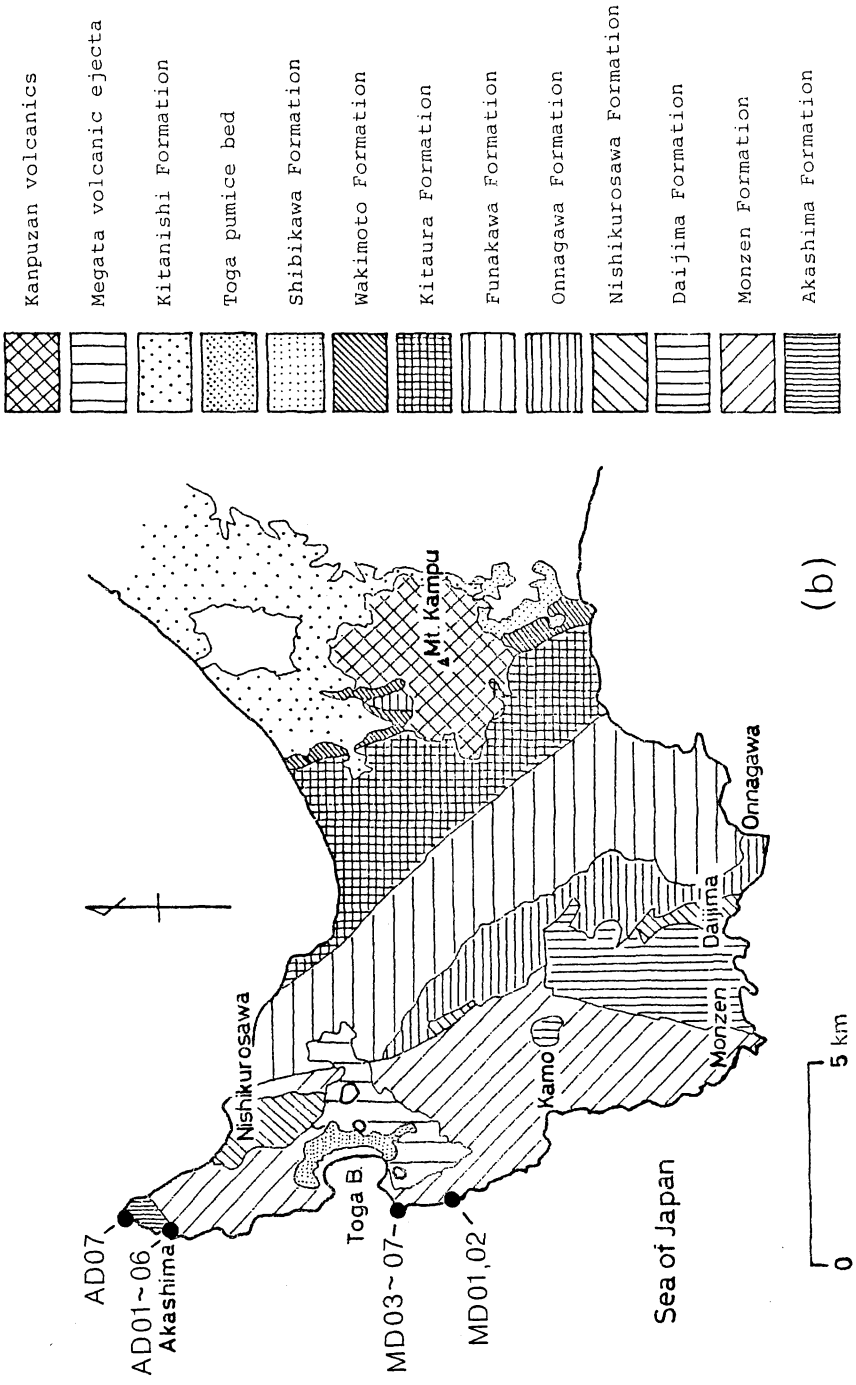

Kanpuzan volcanics
Megata volcanic ejecta
Kitanishi Formation
Toga pumice bed
Shibikawa Formation
Wakimoto Formation
Kitaura Formation
Funakawa Formation
Onnagawa Formation
Nishikurosawa Formation
Daijima Formation
Monzen Formation
Akashima Formation
Mt. Kampu
Nishikurosawa
Toga B.
Kamo
Daijima
Monzen
Onnagawa
AD07
AD01~06
Akashima
MD03~07
MD01,02
Sea of Japan
0
5 km
(b)

Fig. 1.

(about 52 Ma BP), the Monzen Formation (about 35 Ma BP), and the Daijima Formation (about 22 Ma BP). Radiometric ages with the different methods (K–Ar and fission track) were compiled by KANEOKA (1983) and a mean value of the reported ages on each formation is given within parentheses in this text. In the middle and eastern parts of the peninsula, volcanic rocks are covered with later formations (e.g., the Nishikurosawa Formation, the Onnagawa Formation) which are mainly composed of sedimentary rocks. Many dikes are found in the peninsula, especially in the western part. However, very little had been stated about these dike rocks (e.g., TAKAYASU and MATOBA, 1976). Basaltic (doleritic), andesitic, and rhyolitic (dacitic) dikes are well exposed, which strike northeast and dip almost vertically. Only one radiometric age determination by fission track method was reported on an acidic dike at the Akashima Formation and indicated the pre-Miocene intrusion of about 30 Ma BP (SUZUKI, 1980).

3. Measurements and Results

Samples were collected from 14 independent dikes. Seven dikes were emplaced in the Akashima Formation and the others the Monzen Formation. Sampling sites are shown in Fig. 1(b) and their locations and rock types are listed in Table 1. Rocks were

Table 1. Location of the sampling site and rock types and strikes of the dikes.

Site	Location		Rock type	Strike
	Lat.	Lon.		
AD01	40°00′N	139°42′E	Basaltic	N45°E
AD02	40°00′N	139°42′E	Rhyolitic	N42°E
AD03	40°00′N	139°42′E	Rhyolitic	N48°E
AD04	40°00′N	139°42′E	Rhyolitic	N48°E
AD05	40°00′N	139°42′E	Basaltic	N40°E
AD06	40°00′N	139°42′E	Basaltic	N40°E
AD07	40°01′N	139°43′E	Dacitic	N40°E

Site	Location		Rock type	Strike
	Lat.	Lon.		
MD01	39°56′N	139°43′E	Basaltic	N45°E
MD02	39°56′N	139°43′E	Rhyolitic	N45°E
MD03	39°57′N	139°43′E	Andesitic	N40°E
MD04	39°57′N	139°43′E	Andesitic	N45°E
MD05	39°57′N	139°43′E	Andesitic	N50°E
MD06	39°57′N	139°43′E	Andesitic	N40°E
MD07	39°57′N	139°43′E	Andesitic	N53°E

usually sampled with a portable drilling apparatus. In several sites, six or more orientated blocks were collected. One specimen was drilled and cut out from each orientated block in the laboratory. Sixty-two specimens were prepared to measure the remanence by a Schonstedt spinner magnetometer SSM-1A. Specimens were subjected to progressive alternating field demagnetization (AFD) in order to isolate stable components of the remanences from softer secondary ones. For several sampling sites stepwise thermal demagnetization (THD) was also performed on one specimen to ascertain no significant difference between the two cleaning techniques. The progressive cleaning was carried out until the measured remanence was the order of 10% of the initial intensity or until the peak AF reached the maximum one of our demagnetizor (about 200 mT). The direction of the stable magnetization was usually determined by fitting linear lines on the orthogonal vector diagrams (ZIJDERVELD, 1967).

The directions of the stable component of all the specimens are shown in Figs. 2(a) and (b) and a mean direction within each site (site-mean direction) is listed in Table 2 with statistical parameters (FISHER, 1953). In some specimens, the stable components are so hard magnetically that the remanences hardly change their directions and intensities after the removal of the soft components (Fig. 3(b)) or the soft components are so large that only a few percent of the initial remanence remains (Fig. 3(d)). However, the directions of the stable components are compatible with those of other samples from the same site, for which the linear fitting on the orthogonal vector plots was possible (Figs. 3(a) and (c)). Hence, for these sites marked by asterisks in Table 2, the stable component was determined by the optimum demagnetizing field (ODF) which minimized the angular standard variation among the specimens for each site. Mean directions determined at ODFs did not show a good concentration. ODFs are also listed in the rightmost column in Table 2.

In this study, a weak relationship between rock types of dikes and polarities of the magnetization was recognized with one exception; that is, basic (mafic) rocks have reverse magnetizations, intermediate rocks show normal and reverse magnetizations, and acidic (felsic) rocks are magnetized normally. This relationship may reflect the time sequence of the volcanic cycle of the Oga dike swarm. However, more data are needed to discuss the order of the intrusives.

At the site AD07, the directions of stable NRM components are clearly different from those of the other dikes (Fig. 2(a)). We believe that they represent one of the field transition stage from reverse to normal or from normal to reverse because two polarities of the magnetization were observed at the other sites. The mean direction of NRM at the site is excluded in the following discussions. Site-mean directions are shown in Fig. 4(a) by two symbols representing the different host formations, the Akashima and Monzen Formations. The mean direction between sites (formation-mean direction) of dikes at each formation is also listed in Table 2.

4. Discussion

We will first examine the obtained paleomagnetic results. In the following discussion, directions from reversely magnetized specimens were converted to their antipodal positions. The mean inclination, declination, and semi-angle of 95% confidence (α_{95}) of the dikes at the Akashima Formation were calculated to be 58.8°, 8.3°, and 14.0°, respectively. The formation-mean direction was computed to be 57.2° in inclination, −11.8° in declination, and 10.1° in α_{95} on the dikes at the Monzen Formation (the upper figure in Fig. 4(b)). Though a few declination shift was observed, the mean declinations for the two formations coincided with each other within the 95% level of confidence. We calculated mean directions of the normally and the reversely magnetized groups (the lower figure in Fig. 4(b)). The mean inclination, declination, and α_{95} computed to be 57.7°, −14.4°, and 9.9° for the normally magnetized group and they are 58.8°, 10.2°, and 13.6° for the reversely magnetized group. The difference of NRM directions was estimated not to be significant because 95% confidence circles overlapped each other. Based on the above calculation and the fact that both the normally and the reversely magnetized dikes were observed in the two formations, we concluded no systematic difference among the directions of the present dikes.

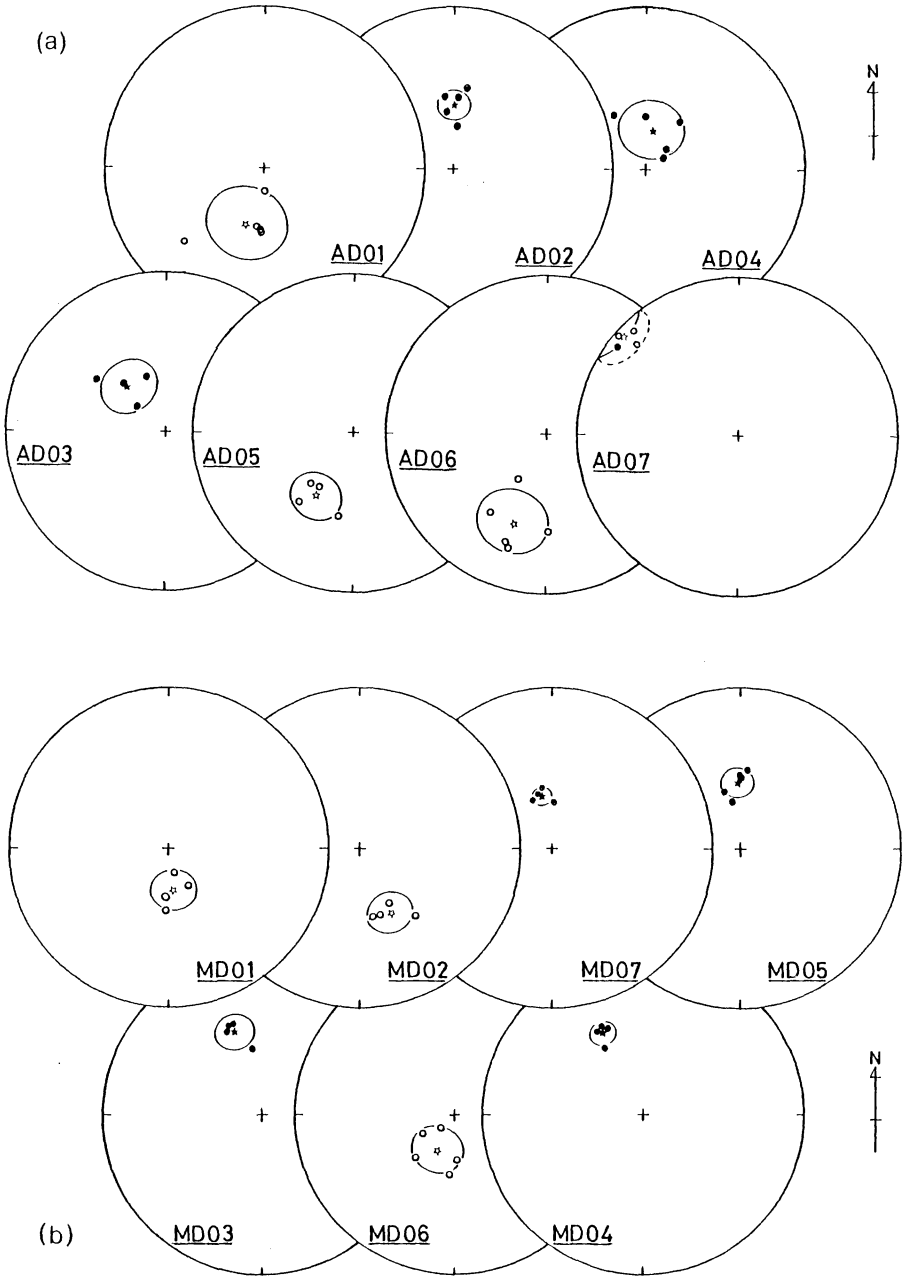


Fig. 2. Equal area projection for stable components of remanence of the Oga dikes intruding into (a) the Akashima Formation and (b) the Monzen Formation. Site mean directions are shown as asterisks with 95% confidence circles. Solid and open symbols correspond to the magnetization vectors in the lower and the upper hemispheres, respectively.

Table 2. Paleomagnetic results for the Oga dikes.

Site	N	Cleaning method	Field direction				Pole position		
			Inc.	Dec.	k	α_{95}	Lat.(N)	Lon.(E)	A_{95}
AD01	5	AFD+THD	-59.7°	-162.9°	16.0	19.7°	-77.0°	42.0°	*(15.0 mT)
AD02	5	AFD+THD	58.5°	1.6°	84.7	8.4°	88.5°	-97.9°	
AD03	4	AFD	60.8°	-42.2°	33.7	16.1°	58.4°	67.3°	*(17.5 mT)
AD04	5	AFD+THD	70.2°	10.4°	24.7	15.7°	74.1°	162.7°	
AD05	4	AFD	-53.2°	-151.7°	45.9	13.7°	-66.6°	56.5°	*(12.5 mT)
AD06	5	AFD+THD	-38.4°	-161.3°	16.7	19.3°	-65.7°	93.6°	*(10.0 mT)
AD07	4	AFD	-5.5°	-48.4°	55.0	12.5°	28.5°	18.2°	
mean	6	(except AD07)	58.8°	8.3°	23.8	14.0°			
MD01	4	AFD	-68.6°	172.7°	62.0	11.8°	-77.1°	-60.5°	
MD02	4	AFD	-53.7°	154.4°	67.1	11.3°	-70.4°	-123.5°	*(100.0 mT)
MD03	4	AFD	45.8°	-18.8°	58.0	12.2°	69.6°	12.8°	
MD04	4	AFD	42.9°	-26.1°	221.2	6.2°	63.6°	23.0°	
MD05	5	AFD+THD	56.6°	-2.2°	101.2	7.6°	86.6°	-8.7°	
MD06	5	AFD+THD	-62.6°	-154.2°	34.5	13.2°	-67.9°	3.3°	
MD07	4	AFD	62.6°	-10.7°	239.4	6.0°	81.1°	79.3°	
mean	7		57.2°	-11.8°	40.0	10.1°			
MEAN	13	(except AD07)	58.3°	-2.8°	28.1	8.0°	77.1°	10.4°	

N denotes the number of specimens used to compute the mean; AFD and THD denote the alternation field demagnetization and the thermal demagnetization; Inc., Dec., and α_{95} are inclination, declination, and semi-angle of the cone of 95% confidence in degrees, respectively. Pole position corresponded to each mean field direction is also shown. Each asterisk at the right column indicates that the mean direction is decided in the optimum demagnetizing field shown in parentheses (see text). The mean field direction at each formation is also presented. The data in MEAN line indicate the mean values of all dikes.

An angular standard deviation (ASD) of 18.1° for the poles with a 95% confidence interval between 14.4° and 25.4° was derived from the between-site precisions on the 13 dikes (WATSON and IRVING, 1957; COX, 1969, 1970). The combined Brunhes data show that at the site between 31° and 45° in latitude the ASD is expected to be 13.9° to 15.8° (MCELHINNY and MERRILL, 1975). The ASD on the Oga dikes was a little larger than that expected but the 95% confidence intervals overlapped each other. Moreover the two polarities of magnetization were observed in the dikes. Then we may estimate roughly that the paleomagnetic data in this study cover the geomagnetic field in the time interval longer than 10^5 yr (e.g., MCFADDEN and MERRILL, 1984). We can conclude that the paleomagnetic results averaged out the effect of the secular variations.

So far, the age of the dike swarm has not been well defined because only one radiometric age determination is available. However, we will here estimate it to be the Middle Miocene or later from the geological and the paleomagnetical point of view presented below.

Miocene basaltic and rhyolitic volcanism are often found in NE Honshu Island. KONDA (1974) pointed out that intensive bimodal volcanism occurred in the Inner Zone of NE Honshu Island in the Middle Miocene, the Nishikurosawa stage (about

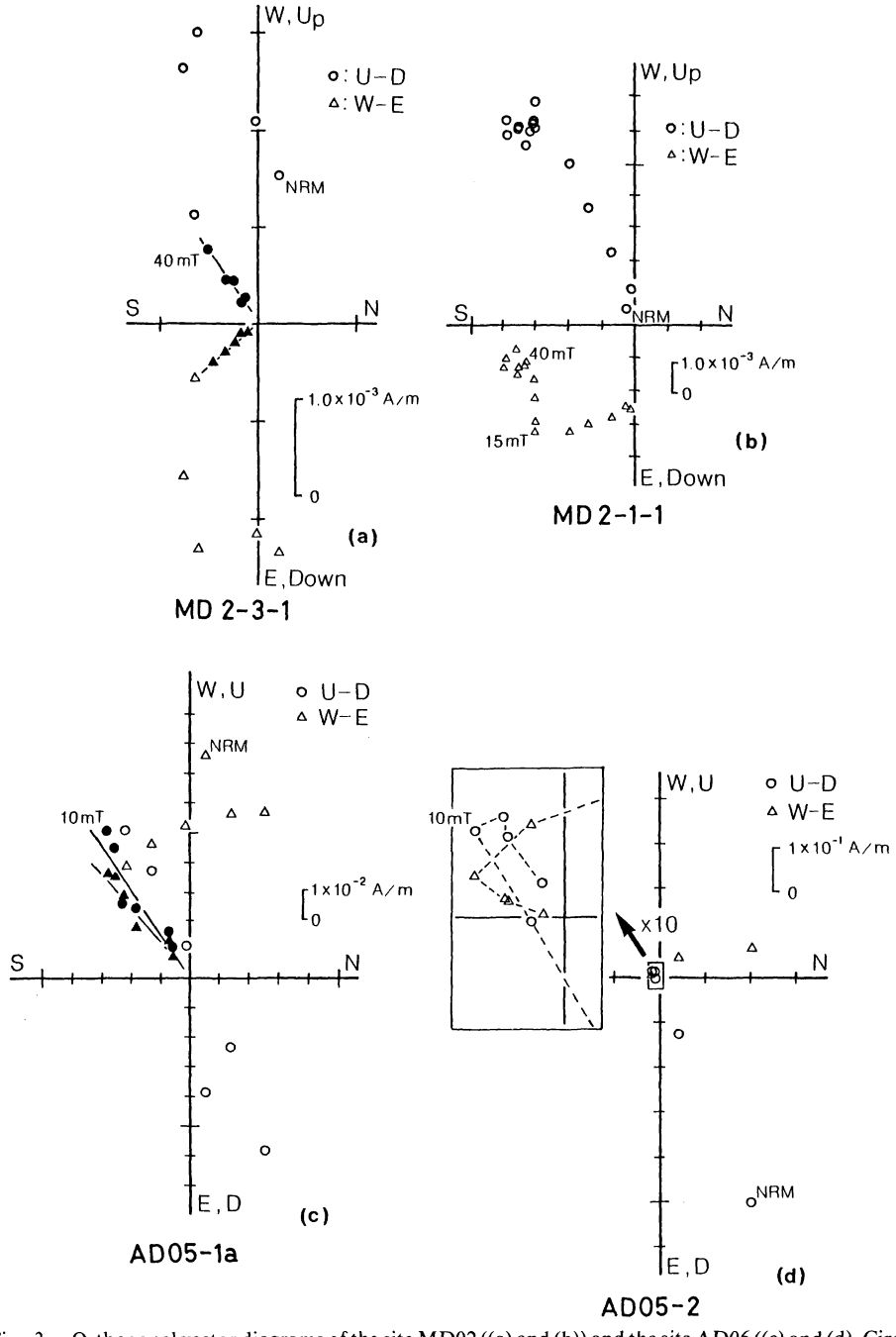


Fig. 3. Orthogonal vector diagrams of the site MD02 ((a) and (b)) and the site AD06 ((c) and (d), Circles and triangles represent the projections of remanence vector end-points on Up- Down (U-D) plane and West-East (W-E) plane including North- South (N-S) axis, respectively. Closed symbols represent the end-points on which straight lines were fitted. The linear fittings were possible in (a) and (c) but were impossible in (b) and (d).

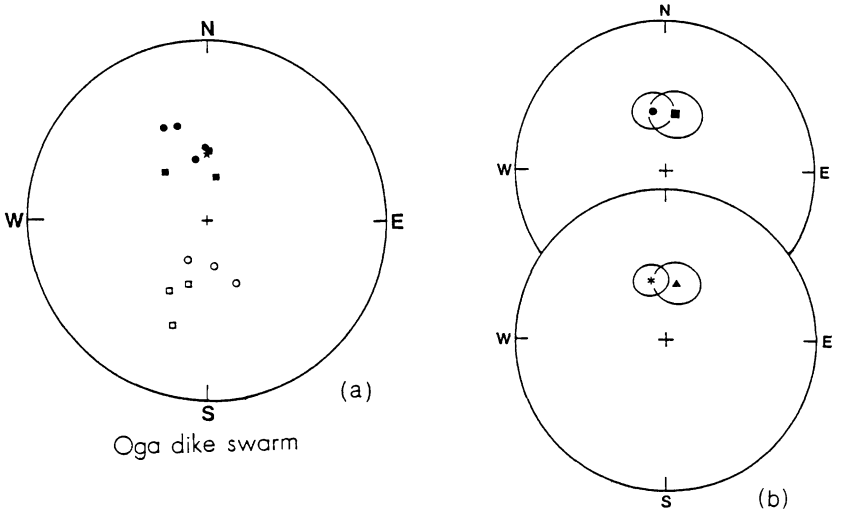


Fig. 4. (a) Site mean directions for thirteen dikes on the equal area projection. Squares and circles represent directions for the dikes at the Akashima and the Monzen Formations, respectively. The geomagnetic field direction expected from the geocentric axial dipole is also shown by an asterisk. Other symbols are the same as those in Fig. 2. (b) (Upper figure) Mean directions for six sites at the Akashima Formation (square) and those for seven sites at the Monzen Formation (circle) with 95% confidence circles. (Lower figure) Star and triangle represent mean directions for seven site-mean directions which were normally magnetized and those for six site-mean direcitos reversely magnetized, respectively. Reversely magnetized directions were converted to their antipodal positions.

15 Ma BP) and also argued that the bimodal volcanism with intermediate facies continued its activity until the Onnagawa stage (about 12 Ma BP and later). Dikes in this study have bimodal rock types with andesitic and dacitic facies and are exposed with almost the same directions of strikes. The remanent magnetizations of stable components from the 13 dikes have northward declinations (mean Inc. $=58.3°$, Dec. $=-2.8°$, see Fig. 3). TOSHA and HAMANO (1986) performed a paleomagnetic study on the formations in the Oga Peninsula and showed that the declinations of remanence at the Monzen and Daijima Formations are northwestward while those at the Nishikurosawa and Onnagawa Formations are northward. The corresponding pole position for the latter formations was close to that of the present Oga dikes. These pole positions and the recent results for NE Honshu Island by other investigators are compiled in Fig. 5. OTOFUJI *et al.* (1985) presented two pole positions. One was of 11 to 14 Ma BP represented by No. 2 in Fig. 5 and the other ranging between 20 and 32 Ma BP by No. 1. TSUNAKAWA *et al.* (1985) performed paleomagnetism on the dikes with N45° E to N60° E strikes and presented K–Ar age determination of about 8 Ma BP. Their paleomagnetic results are also plotted in Fig. 5 (No. 3). The pole position of the Oga dikes in this study was far from those in pre-Middle Miocene age (Capital letters M, D, and No. 1 in Fig. 5) but was close to those in the Middle Miocene (Capital letter N, No. 2, and No. 3 in Fig. 5). Therefore, the dikes were presumed to intrude at Middle Miocene age or later.

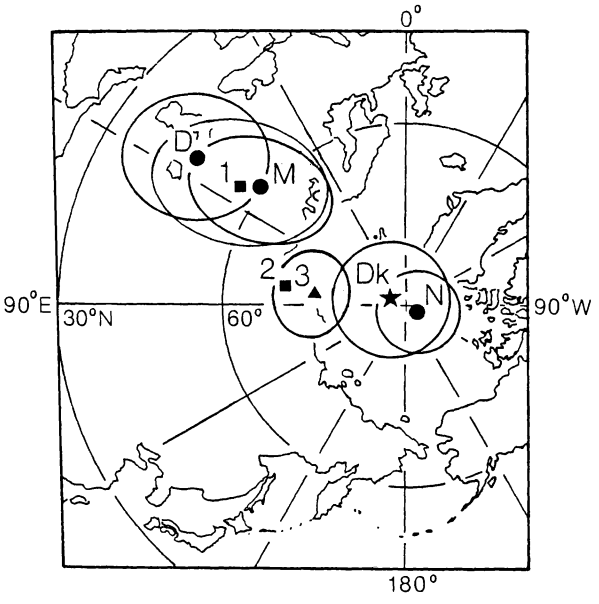


Fig. 5. Mean pole positions in the Oga Peninsula and in NE Honshu Island. Circles represent those from the formations in the peninsula after TOSHA and HAMANO (1986) (M—Monzen Formation, D—Daijima Formation, N—Nishikurosawa and Onnagawa Formations). An asterisk (Dk) shows that from the dikes in this study. Squares (after OTOFUJI *et al.* (1985)) and a triangle (after TSUNAKAWA *et al.* (1985)) indicate pole positions in the northern part of NE Honshu Island (No. 1—11 to 14 Ma BP, No. 2—20 to 32 Ma BP, No. 3—about 8 Ma BP).

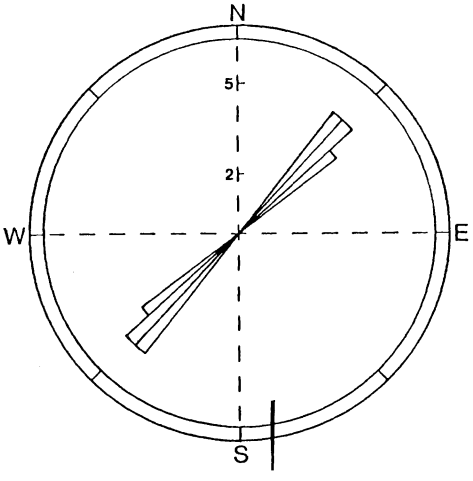


Fig. 6. Rose diagram of 14 dikes in this study.

We can estimate the direction of the horizontal maximum stress field, σ_{Hmax}, from the dominant direction of strikes in a dike swarm (NAKAMURA, 1977). Rose diagram of strikes for the Oga dikes indicates the dominant strike as a NE direction (Fig. 6). The paleomagnetic field directions of the dikes indicate that they intruded after NE Honshu Island had rotated counterclockwisely. Therefore, it is inferred that the σ_{Hmax} direction of NE represented the stress field after the tectonic rotation had completed. TSUNAKAWA (1986) presented the temporal stress field change in NE Honshu Island. However, his investigation area was the southern part of NE Honshu Island and a NE trend of σ_{Hmax} was not shown. SATO *et al.* (1982) compiled the strikes of the intruded dikes and the veins of ore deposits in the northern part of NE Honshu Island in Miocene time. In their opinion, dike swarms with NE strikes are found at various areas in NE Honshu Island (See Fig. 1) and their intrusive ages were geologically estimated to be the Middle to Late Miocene. HORIKOSHI (1983) compiled the distribution of strikes of the Neogene vein deposits associated with Kuroko ore deposits in NE Japan. Vein deposits with NE–SW orientation are predominantly distributed in the Akita–Iwate zone, the northern part of NE Honshu Island and were probably formed in Middle to Late Miocene time. Hence, the NE direction of σ_{Hmax} was not a special case in the Oga Peninsula but one of general trends in the northern part of NE Honshu Island during Miocene age.

Stress field in Early Miocene time was estimated to be different from those in the Middle and Late Miocene (SATO *et al.*, 1982). The present study indicates that the σ_{Hmax} direction of the Middle to Late Miocene was the stress field posterior to the tectonic rotation. Since the rotation of NE Honshu Island occurred during Early Miocene age (TOSHA and HAMANO, 1986; OTOFUJI *et al.*, 1985), the stress field in the Early Miocene can be related to the counterclockwise rotation of NE Honshu Island.

5. Conclusion

From the paleomagnetic study of the dikes in the Oga Peninsula, we conclude that these dikes intruded in Middle Miocene or later ages after the completion of tectonic rotation of NE Honshu Island. The NE direction of σ_{Hmax} inferred from the orientation of the dikes, therefore, represents the stress field after the rotation had completed.

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